

BREAKHOOK · DESIGN DECISION RECORD

BreakHook design for construction

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DOCUMENT CONTROL

Document ID	BHK-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breakhook		
Source	docs/decisions/0002-design-for-construction.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Concept made constructable (STANDARDS section 18), under Amish's 2026-10-03 pre-approval

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** accepted
- **Decided by:** Amish Chadha, by pre-approval: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." Standing instruction (2026-09-30): "fix the design assumptions to match and be physically feasible".

Safety: These changes keep every safety feature of the concept: the pole is withdrawn before the haul, there is no metal in the insulated length, and every part in the rope load path is rated and proof-loaded. None of them changes what the product does or its pitch.

■ Context

The TRL 2 massing model showed a hook, a pole and a rope in the right proportions but left open how each part is made and how it joins its neighbours. A constructability review of `cad/src/model.py` checked every pair of parts for overlap (none over 1 mm³ after the changes), every fit (clearances printed by `python cad/src/model.py --check`) and the order of assembly.

■ Changes

Component	The concept had	The constructable design has	Why
Hook head	A forged or fabricated hook with a spike	One profile cut from 10 mm S355 plate (spike, 36 mm shank, raked arm, rope tab)	No forging; a local welder can cut it by plasma or by drilling and grinding
Hook to pole	Hook on the end of the pole	A 50.8 x 2.0 steel socket, slotted 10.5 x 50 across one end; the plate's 56 mm wide root sits in both slots and stands 2.6 mm proud for four 6 mm fillet welds	Face-to-face, welded joint that a welder can check by eye
Pole in socket	Not defined	Slip fit, 1.15 mm radial clearance, 130 mm deep, stopping on the plate; a rubber tape ring gives a 50 N release	The pole must come off before the haul (BHK-DDR-001, decision 2)
Pole joints	Not defined	External fibreglass sleeve 50.8 x 3.2 x 300, bored to a slip fit, bonded 150 mm to the lower section; nylon 10 mm pin 75 mm above the joint	Foam-filled tube cannot take an internal spigot; nylon keeps metal out of the insulated length
Rope attachment	Pull rope tied to the hook head	13 mm hole in a rope tab in line with the arm; rated bow shackle; 1.5 m wire rope leader; second shackle to the rope's spliced eye	A knot is weaker and can be cut by sheet edges; the line through the arm keeps the shank in tension
Hook plate	40 mm shank, 40 mm arm, S275	36 mm shank, 48 mm arm, S355	The arm on a 100 mm beam was at 1.24 on yield at proof with S275; the narrower shank pays for the wider arm and keeps R5
Hauling handles	Handles or toggles	300 mm fibreglass toggles on 6 mm prusik cord loops, 1 m apart from 11 m	Movable on the rope, nothing to cut into the rope, and no metal
Pole ends and markings	Not defined	Rubber butt cap; epoxy-sealed ends and pin holes; red hand band 1.55 m from the butt	Keeps water out of the foam; tells the pole team where to hold
Storage rack	Rack and seal	Two welded uprights of 40 x 6 flat bar with lipped arms and a rope peg; poles 56 mm apart with sleeves at alternate ends; coated lock cable, padlock, numbered seal	Every piece rests on an arm and nothing touches its neighbour

■ Consequences

- Mass rises to 7.82 kg for the pole and hook head, inside the 8 kg limit (BHK-CAL-001 [F10]).
- The model, STEP and STL exports, the general arrangement BHK-DWG-001 Rev P1, the making sketches BHK-DWG-101 to 106 and the concept media were regenerated from the constructable model.
- `design_state: constructable` is set in `project.yaml`.
- Fits that depend on bought parts (sleeve bore, tube tolerance, shackle jaw) are listed under "To confirm when parts are bought" in BHK-DEC-001.